

LOW CODE

Zoho Creator

Ex. No: 3

Name: Abishek D
Roll No: 240701013

APPLICATION BUILDING USING ZIA

AIM

To build a “Student Crop Advisory System” application in Zoho Creator using Zia App Builder by giving a natural-language prompt, to review and refine the forms, reports, dashboard and workflow generated by Zia, and to test the application by submitting and viewing sample records.

SOFTWARE / TOOLS USED

- Zoho Creator (Low-Code Application Development Platform) with Zia enabled
- Zia App Builder (AI-based application generator in Zoho Creator)
- Web browser (Google Chrome / Microsoft Edge)

THEORY

Zia is the AI assistant of Zoho. In Zoho Creator, **Zia App Builder** creates a working application from a prompt written in plain language. The user describes the purpose of the application, the data to be collected and the process to be followed, and Zia interprets it using a Large Language Model (LLM) and automatically generates the application components such as forms and fields, reports, dashboard pages, workflows and user permissions.

Zia is configured once under **Operations** → **Zia** by selecting an LLM provider and enabling the **Create Application** feature. Zia App Builder works in two modes: **Build mode**, which generates the application directly from the prompt, and **Plan mode**, which first prepares a Business Requirement Document (BRD) for the user to review before the application is built. After generation, the application can be opened in edit mode and refined with the normal drag-and-drop builder, or with further Zia assistance such as field suggestions. The more detailed the prompt, the closer the generated application is to the requirement.

A **Student Crop Advisory System** helps agriculture students to submit queries about a crop (season, soil type, land area, problem) and to receive an advisory on fertilizer, irrigation and pest control from an advisor. This is a good use case for Zia because it needs several related forms, reports and a simple workflow.

PROMPT GIVEN TO ZIA

Create an application named "Student Crop Advisory System". Students of an agriculture college submit queries about a crop by entering their name, email, year of study, crop, season, soil type, land area, problem description and a crop photo. Add a Crop Master form to store crop name, category, season, soil type, water requirement, recommended fertilizer and common pests and diseases. An advisor reviews each query and records an advisory with fertilizer, irrigation and pest control recommendations. The query status should be Pending when submitted and should change to Answered once the advisory is added, and the student should be notified by email. Provide reports for crops, student queries and advisories, and a dashboard showing the number of queries by status and by crop.

APPLICATION COMPONENTS

The application “Student Crop Advisory System” generated by Zia contains the following forms and fields (field names may vary slightly in the generated application):

Form	Field Name	Field Type	Description
Crop_Master	Crop_Name	Single Line	Name of the crop
	Crop_Category	Dropdown	Cereal / Pulse / Vegetable / Oilseed / Cash Crop
	Season	Dropdown	Kharif / Rabi / Zaid
	Soil_Type	Dropdown	Red / Black / Alluvial / Sandy / Loamy
	Water_Requirement	Dropdown	Low / Medium / High
	Recommended_Fertilizer	Multi Line	Standard fertilizer recommendation for the crop
	Common_Pests_Diseases	Multi Line	Common pests and diseases of the crop
Student_Query	Student_Name	Name	Full name of the student
	Email	Email	Email address for the advisory notification
	Year_of_Study	Dropdown	I / II / III / IV
	Crop	Lookup	Crop selected from Crop_Master
	Season	Dropdown	Kharif / Rabi / Zaid
	Soil_Type	Dropdown	Type of soil in the field
	Land_Area_Acres	Decimal	Area of the land in acres
	Problem_Description	Multi Line	Problem faced in the crop
	Crop_Photo	Image / File Upload	Photo of the affected crop
	Status	Dropdown	Pending / Answered (default: Pending)

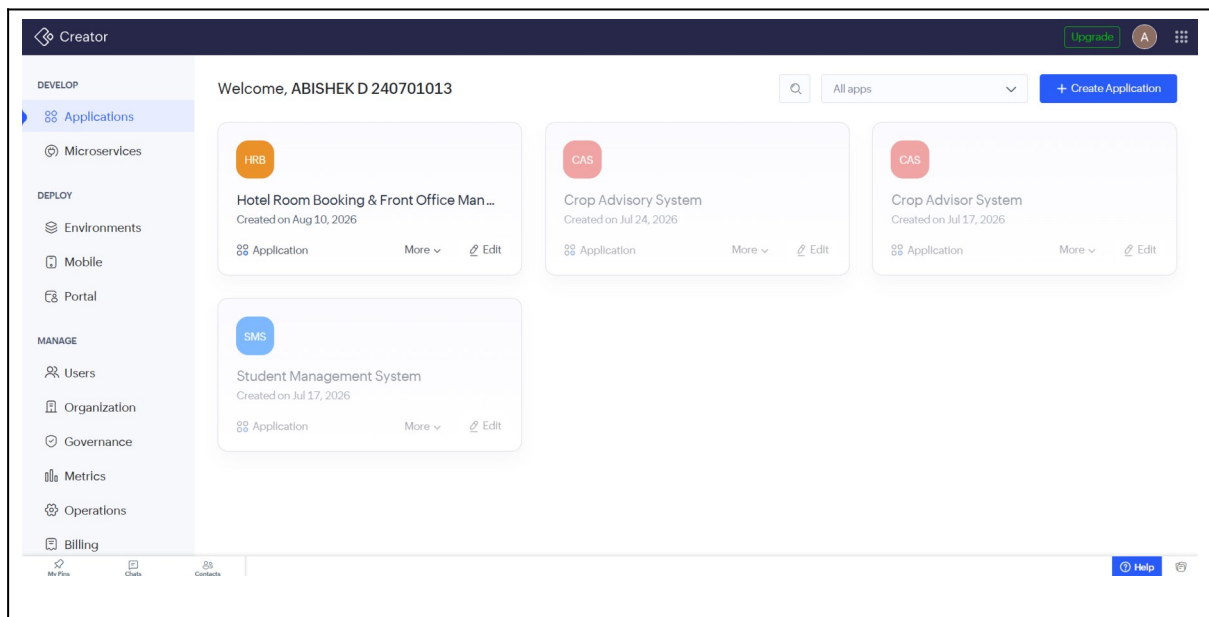
Form	Field Name	Field Type	Description
Advisory	Query	Lookup	Student query being answered
	Advisor_Name	Name	Name of the advisor
	Fertilizer_Advice	Multi Line	Fertilizer type, quantity and timing
	Irrigation_Advice	Multi Line	Irrigation schedule
	Pest_Disease_Control	Multi Line	Pest and disease control measures
	Advisory_Date	Date	Date of the advisory

Other components generated: **Reports** – Crop_Master Report, Student_Query Report and Advisory Report; **Page** – Advisory Dashboard (queries by status and by crop); **Workflow** – on submission of an Advisory, the Status of the query is changed to Answered and an email is sent to the student.

PROCEDURE

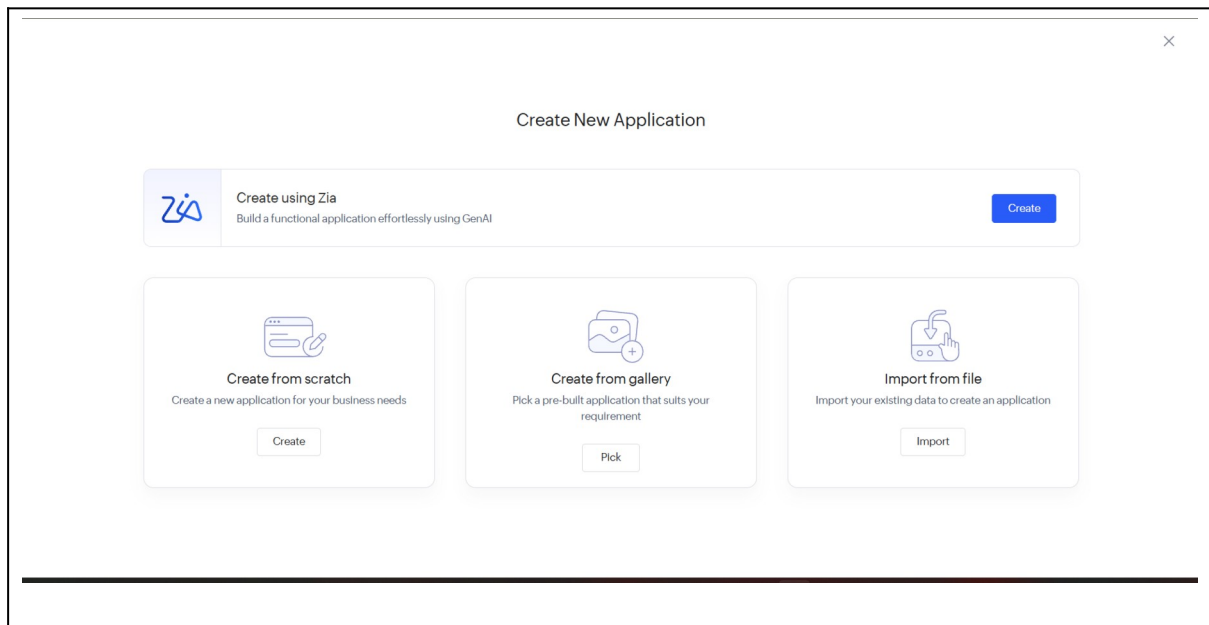
- **Sign in to Zoho Creator:** Open Zoho Creator in the web browser and sign in with the Zoho account. The dashboard opens on the Solutions page.

Screenshot 1: Zoho Creator dashboard



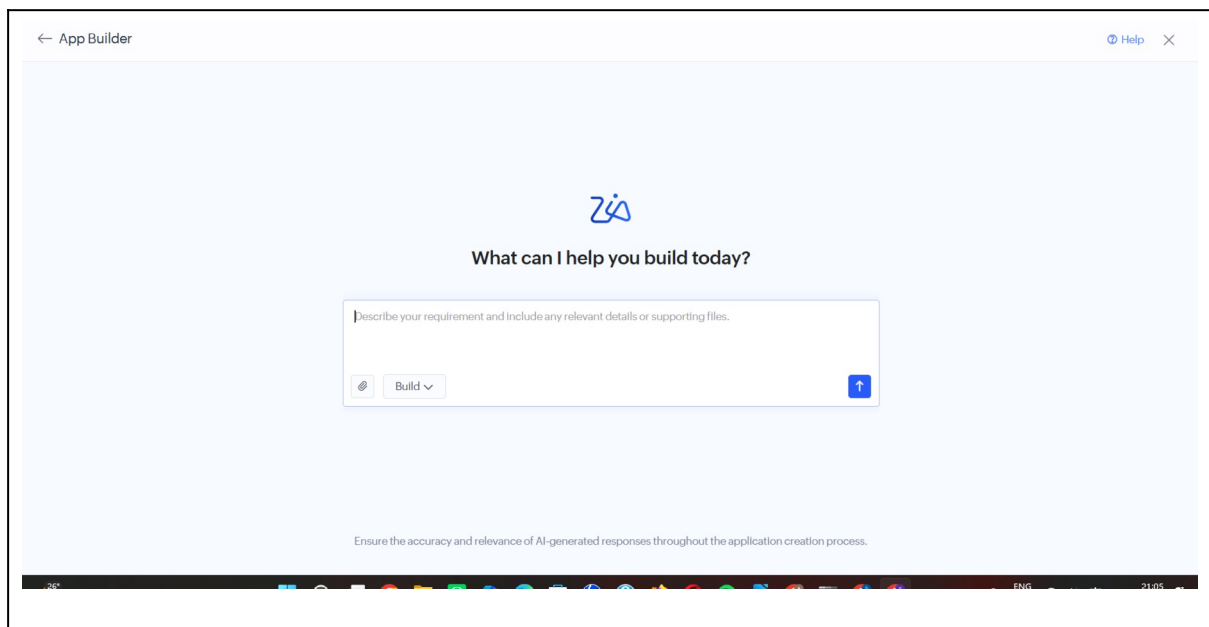
- **Enable Zia (one-time setup):** Go to Operations → Zia, select the LLM provider, and under the Features tab turn on the Access toggle for Create Application. Save the settings.

Screenshot 2: Zia settings with Create Application enabled



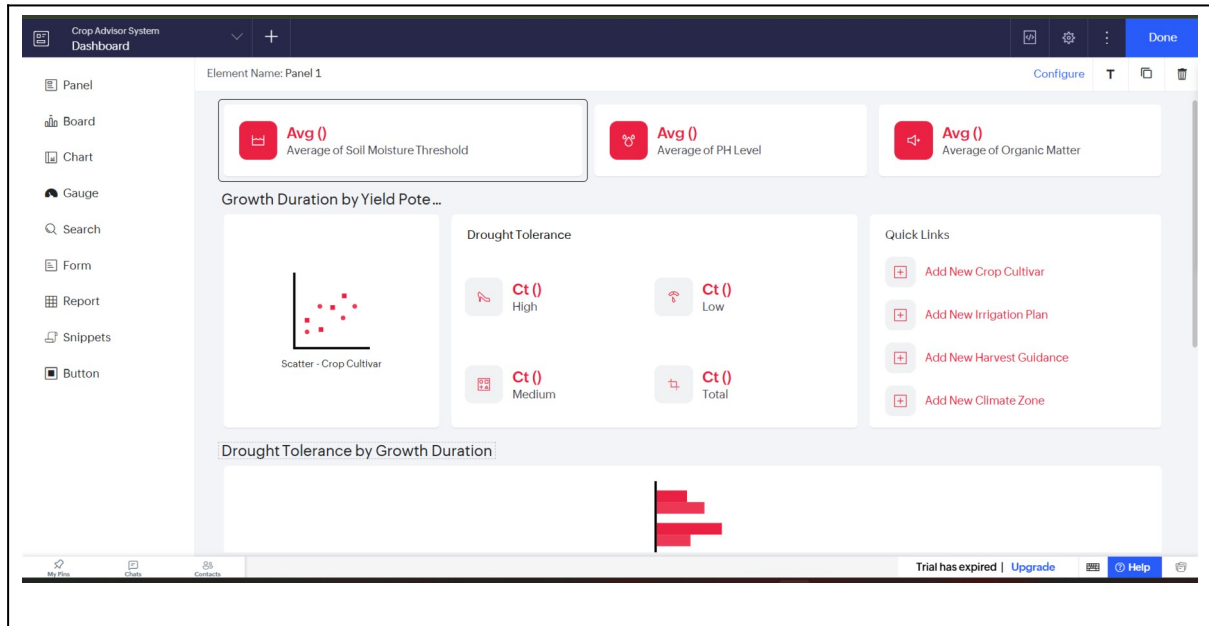
- **Open Zia App Builder:** On the dashboard, click + Create Solution, select the Applications card and choose Create using Zia.

Screenshot 3: Create using Zia option



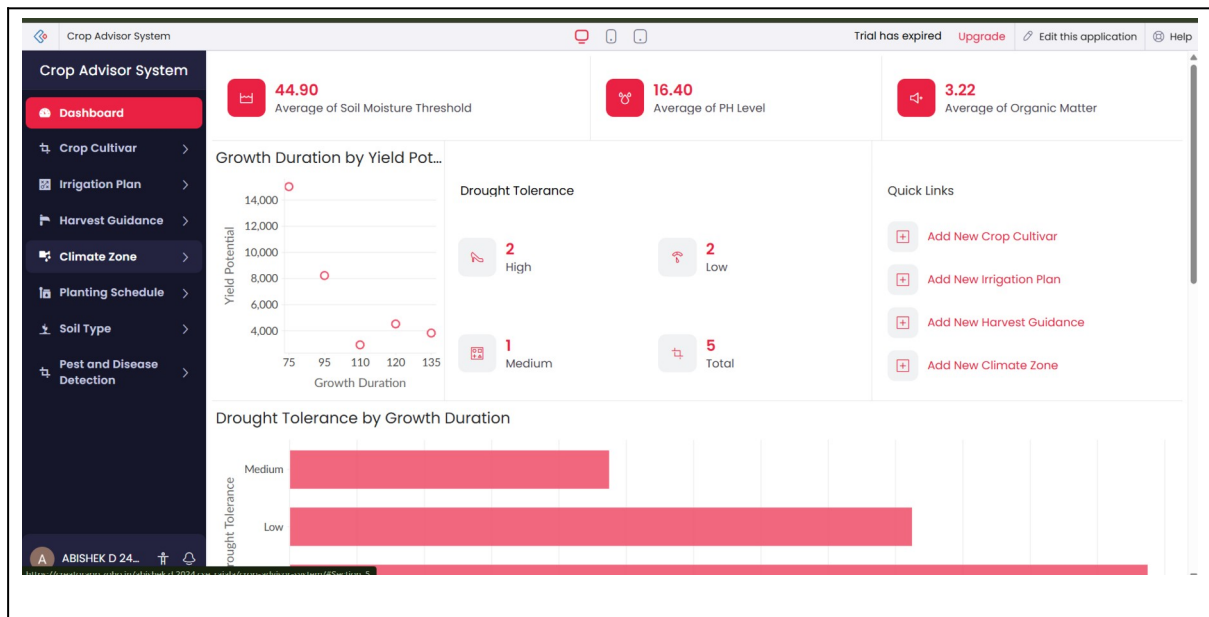
- **Enter the prompt:** Keep the mode as Build (Plan mode can be used to review a BRD first), select the LLM model, type the prompt given above in the prompt box and submit it.
- **Generate the application:** Zia analyses the prompt and builds the forms, reports, dashboard and workflow. Wait until the application is generated and opens.

Screenshot 5: Zia generating the Student Crop Advisory System



- **Review the forms:** Open the application in edit mode and check that the Crop_Master, Student_Query and Advisory forms are created with the required fields and field types.
- **Refine the application:** If a field is missing or wrongly named (for example Land_Area_Acres or Crop_Photo), add or edit it in the form builder. Zia field suggestions or Add component → Form → Using Zia can also be used. Set Status as a Dropdown with the default value Pending.

Screenshot 7:



- **Review the reports and dashboard:** Check the Crop_Master, Student_Query and Advisory reports and the Advisory Dashboard page created by Zia.

Screenshot 8: Reports and dashboard generated by Zia

The screenshot shows the 'Crop Cultivar Report' dashboard. The top navigation bar is identical to Screenshot 7. The left sidebar highlights 'Crop Cultivar Report'. The main content area displays a table with the following data:

Cultivar ID	Crop Name	Variety	Growth Duration	Yield Potential	Drought Tolerance	Disease Resistance	Ideal Soil
5	Tomato	Roma VF	75	15000	Low	Virus	
4	Soybean	Roundup Ready	110	2900	Low	Virus Blight Mosaic	
3	Rice	Basmati 370	135	3800	High	Virus Rot Leaf Spot	
2	Corn	Sweet Gold	95	8200	Medium		
1	Wheat	Hybrid 2025	120	4500	High	Rot Leaf Spot	

The table is paginated, showing 5 of 5 records. The user profile 'ABISHEK D 24...' is visible at the bottom left.

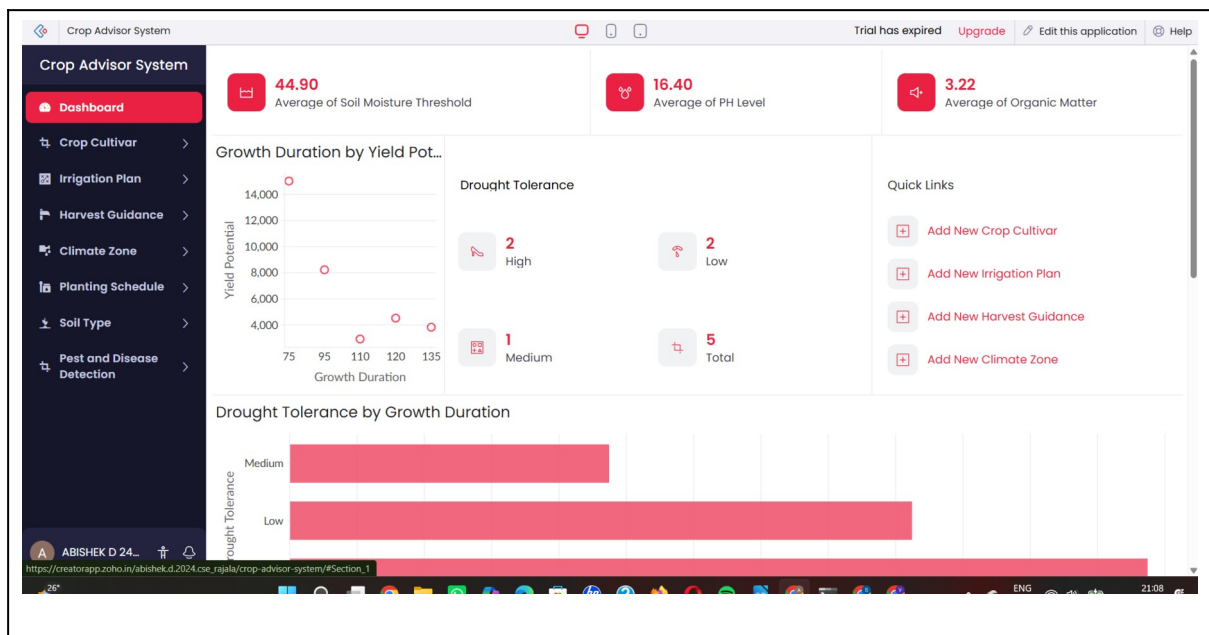
- **Check the workflow:** Open the workflow of the Advisory form and verify that it updates the query Status to Answered and sends an email to the student. If it is not present, create it as a form workflow on successful submission.

Screenshot :

Crop	Symptom Description	Common Pests	Common Diseases	Risk Level	Prevention Tips	Recommended Action
1	3	Spider Mites Thrips	Downy Mildew Leaf Blight	High		
1	3	Whiteflies Beetles Spider Mites Thrips	Powdery Mildew Leaf Blight	Medium		
1	4	Spider Mites Aphids	Root Rot Stem Rot	Low		

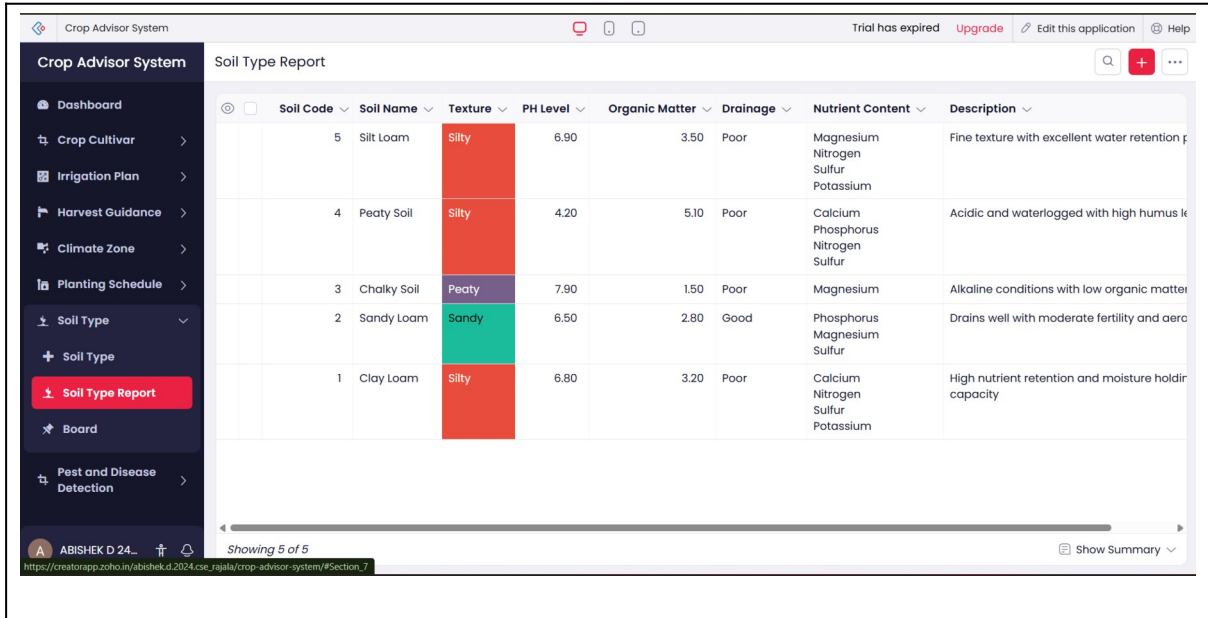
- **Save and run the application:** Click Save and then Access Application to open the application in live mode.

Screenshot 10: Application in live mode



- **Add sample crop data:** Open the Crop_Master form and add three or four crops (for example Rice, Groundnut, Tomato and Cotton) with their details.
- **Submit student queries:** Open the Student_Query form, fill in sample details, select a crop and submit. Repeat for one or two more queries.

Screenshot:



The screenshot displays the 'Crop Advisor System' interface. On the left is a dark sidebar with navigation options: Dashboard, Crop Cultivar, Irrigation Plan, Harvest Guidance, Climate Zone, Planting Schedule, Soil Type (selected), Soil Type, Soil Type Report (highlighted in red), Board, and Pest and Disease Detection. The main area shows a 'Soil Type Report' table with the following data:

Soil Code	Soil Name	Texture	PH Level	Organic Matter	Drainage	Nutrient Content	Description
5	Silt Loam	Silty	6.90	3.50	Poor	Magnesium Nitrogen Sulfur Potassium	Fine texture with excellent water retention p
4	Peaty Soil	Silty	4.20	5.10	Poor	Calcium Phosphorus Nitrogen Sulfur	Acidic and waterlogged with high humus k
3	Chalky Soil	Peaty	7.90	1.50	Poor	Magnesium	Alkaline conditions with low organic matter
2	Sandy Loam	Sandy	6.50	2.80	Good	Phosphorus Magnesium Sulfur	Drains well with moderate fertility and aerc
1	Clay Loam	Silty	6.80	3.20	Poor	Calcium Nitrogen Sulfur Potassium	High nutrient retention and moisture holdi capacity

At the bottom of the table, it says 'Showing 5 of 5' and 'Show Summary'.

- **Add an advisory:** Open the Advisory form, select a pending query, enter the fertilizer, irrigation and pest control advice and submit. Verify that the Status of that query changes to Answered.
- **View the reports and dashboard:** Open the Student_Query Report, Advisory Report and the Advisory Dashboard and verify that the records and charts are displayed correctly.

RESULT

Thus, the application “Student Crop Advisory System” was built in Zoho Creator using Zia App Builder from a natural-language prompt, the generated forms, reports, dashboard and workflow were reviewed and refined, and the submitted student queries and advisories were viewed successfully.